TITLE : LOW-COST SMART SYSTEM FOR HOME SAFETY AND COMFORT USING MACHINE LEARNING WITH TELEMEDICINE AND HUMAN SUPERVISION

Fethi Nedale LATTAB[1],Naima Boukhiar[1], Ghalem Belalem[1]& Amin Bouamar[1]*

1- LIO Laboratory, University of Oran 1, Oran, Algeria

Abstract:

In modern urban living, homes face increasing threats from hazards such as fires, gas leaks, and carbon monoxide poisoning. To address these risks, this paper introduces an intelligent, low-cost home automation system leveraging machine learning for real-time anomaly detection, predictive threat analysis, and health monitoring. The system integrates telemedicine features aimed at improving health monitoring for elderly individuals, including tracking vital signs and emotional states. Additionally, it includes supervision capabilities for individuals in difficulty, providing timely alerts to caregivers, emergency services, and medical professionals. IoT sensors (flame, gas, motion, heart rate, and SpO2) are incorporated into the system, which also features a mobile application for instant alerts, remote monitoring, and emergency response coordination. Key innovations include adaptive energy optimization, automated hazard mitigation, and the capacity to monitor and respond to specific health conditions in elderly residents. Preliminary results show high detection accuracy, robust health supervision capabilities, and rapid response times, positioning this system as a smart solution for enhancing both home safety and health management in urban environments. The ultimate goal is to provide a comprehensive solution that ensures home safety and supports the health and well-being of vulnerable populations.

Keywords : Home safety, IoT , Machine Learning, Mobile application ,Telemedicine ,Elderly supervision